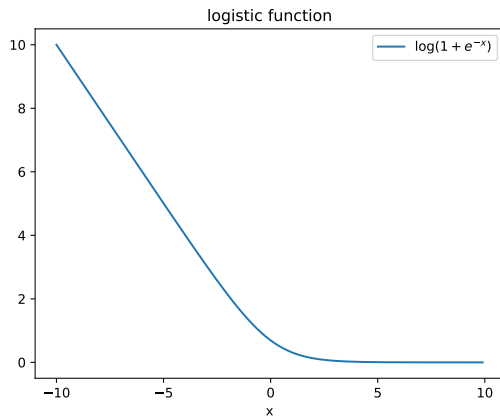


Fondamentaux théoriques du machine learning



Overview of lecture 5

Classification

- Problem statement

- Convexification of the risk and calibration

- Logistic regression

Probabilistic modelling

Classification

- Problem statement

- Convexification of the risk and calibration

- Logistic regression

Probabilistic modelling

General classification problem

- ▶ $\mathcal{X} = \mathbb{R}^d$
- ▶ $\mathcal{Y} = \{-1, 1\}$ or $\mathcal{Y} = \{0, 1\}$.
- ▶ $l(y, z) = 1_{y \neq z}$ ("0-1" loss)
- ▶ $F = \mathcal{Y}^{\mathcal{X}}$

Problem

Optimizing on $F = \mathcal{Y}^{\mathcal{X}}$ is equivalent to optimizing in the set of subsets of $\mathcal{X}$.

We cannot differentiate on this hypothesis space and it is not clear how to regularize.

Subsets

Exercise 1 : Combinatorial problem

If we wanted to try all applications in $\mathcal{Y}^{\mathcal{X}}$, if $|\mathcal{X}| = n$, how many applications would there be ?

Real-valued function

Instead of an application in $\mathcal{Y}^{\mathcal{X}}$, we will learn $g : \mathcal{X} \rightarrow \mathbb{R}$ and define $f(x) = \text{sign}(g(x))$ with

$$\text{sign}(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Risk

The risk (generalization error) of $f = \text{sign} \circ g$ is.

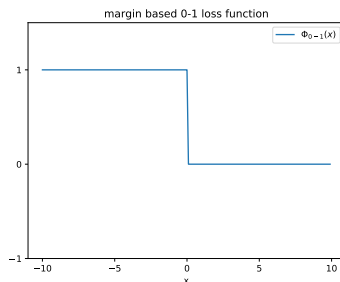
$$\begin{aligned} R(g) &= P(\text{sign}(g(x)) \neq y) \\ &= E \left[1_{\text{sign}(g(x)) \neq y} \right] \\ &= E \left[1_{yg(x) < 0} \right] \end{aligned} \tag{1}$$

Remark : several solutions

If f^* is the Bayes predictor, there might be many optimal functions g , i.e : such that $\text{sign}(g(x)) = f^*(x)$.

Margin based 0-1 loss function Φ_{0-1}

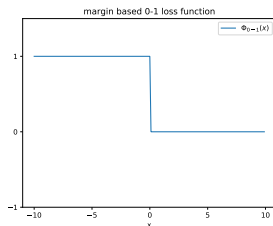
$$\begin{aligned} R(g) &= E \left[1_{\text{sign}(g(x)) \neq y} \right] \\ &= E \left[1_{yg(x) < 0} \right] \\ &= E \left[\Phi_{0-1}(yg(x)) \right] \end{aligned} \tag{2}$$



Empirical risk minimization

The corresponding empirical risk writes :

$$\frac{1}{n} \sum_{i=1}^n \Phi_{0-1}(y_i g(x_i)) \quad (3)$$

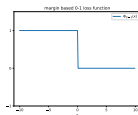


What is the issue with this objective function ?

Empirical risk minimization

The corresponding empirical risk writes :

$$\frac{1}{n} \sum_{i=1}^n \Phi_{0-1}(y_i g(x_i)) \quad (4)$$



What is the issue with this objective function ?

- ▶ non-convex
- ▶ the differential is not defined or non informative.

Convex surrogate

Key idea : replace Φ_{0-1} by another function Φ that is easier to optimize (convexity) but still represents the correctness of the classification.

Definition

The Φ -risk is defined as

$$R_{\Phi}(g) = E \left[\Phi(yg(x)) \right] \quad (5)$$

The empirical Φ -risk is defined as

$$R_{\Phi,n}(g) = \frac{1}{n} \sum_{i=1}^n \Phi(y_i g(x_i)) \quad (6)$$

Most common convex surrogates

Definition

Logistic loss

$$\Phi(u) = \log(1 + e^{-u}) \quad (7)$$

With linear predictors, this loss will lead to **logistic regression** (which is classification despite its name).

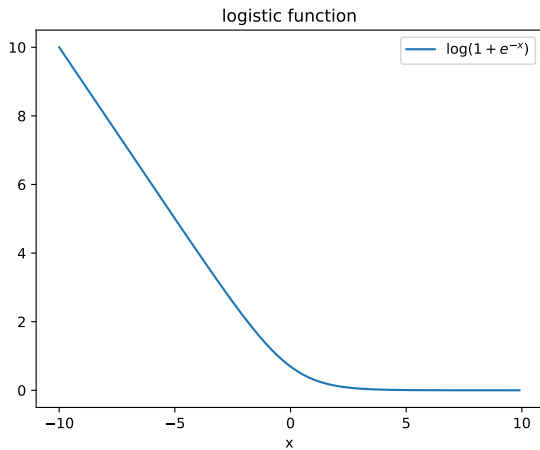
Most common convex surrogates

If $\mathcal{Y} = \{0, 1\}$, $\hat{y}$ is the prediction and y is the correct label, then we write :

$$l(\hat{y}, y) = y \log(1 + e^{-\hat{y}}) + (1 - y) \log(1 + e^{\hat{y}}) \quad (8)$$

(cross entropy loss)

Logistic function



Most common convex surrogates

Definition

Hinge loss

$$\Phi(u) = \max(1 - u, 0) \quad (9)$$

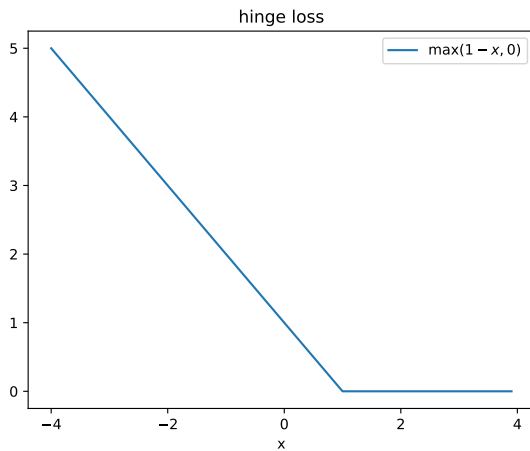
With linear predictors, this loss will lead to **Support vector machines**.

Definition

Squared hinge loss

$$\Phi(u) = (\max(1 - u, 0))^2 \quad (10)$$

Hinge loss

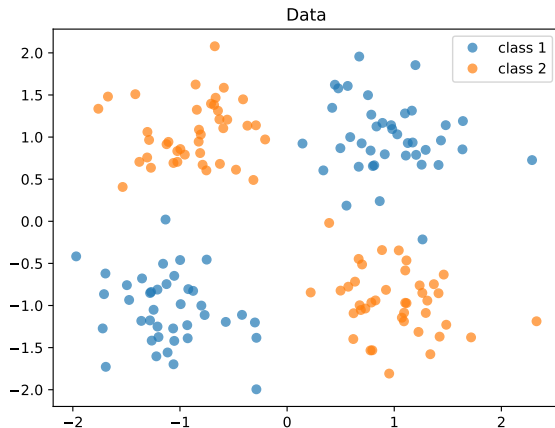


Under some technical hypotheses, minimizing the empirical Φ risk leads to a good generalization error for the "0-1" loss (notion of calibration function).

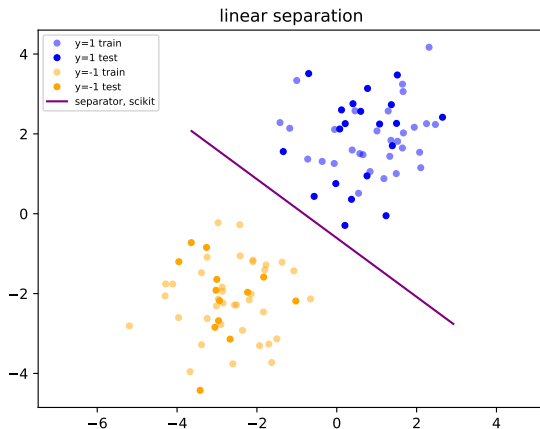
Logistic regression

- ▶ $g(x) = \langle x, \theta \rangle = x^T \theta.$
- ▶ $f(x) = \text{sign}(\langle x^T \theta \rangle)$
- ▶ It can be seen as "linear regression applied to classification".

Data to separate



Data to separate



Logistic regression

In this section we use the setting $\mathcal{Y} = \{0, 1\}$.

- ▶ prediction : $\hat{y} = x^T \theta$
- ▶ surrogate loss : cross-entropy loss.

$$l(\hat{y}, y) = y \log(1 + e^{-\hat{y}}) + (1 - y) \log(1 + e^{\hat{y}}) \quad (11)$$

Logistic regression estimator

If l is the logistic loss, it is defined as

$$\hat{\theta}_{logit} = \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n l(x_i^T \theta, y_i) \quad (12)$$

We will show that this empirical risk is convex as a function of θ (exercises).

No closed-form solution

Since the loss is convex, to minimize it is sufficient to look for the cancellation of the gradient. However, the corresponding equation has no closed-form solution.

We thus need to use iterative algorithms in order to find a minimizer (e.g. : gradient descent, Newton's method, etc)

Practical usage of logistic regression

In practice, it is common to :

- ▶ regularize the logistic loss to avoid overfitting, for instance with a $L2$ penalty (as in ridge regression)
- ▶ use feature maps and classify with $\phi(x)$ instead of x .

Classification

Problem statement

Convexification of the risk and calibration

Logistic regression

Probabilistic modelling

Context

We are given a set of observations $\{y_1, \dots, y_n\} \in \mathcal{Y}$ that we assume are generated i.i.d from an unknown distribution. We look for a **probabilistic model** that explains well the data. We could for instance use this model to generate new data, that would be statistically similar to the observed ones.

Density estimation

We will consider **parametric models** for density estimation.

Definition

Parametric model

Let $\Theta \subset \mathbb{R}^p$. A parametric model $\mathcal{P}$ is a set of probability distributions on $\mathcal{Y}$, indexed by elements of Θ .

$$\mathcal{P} = \{p_\theta | \theta \in \Theta\}$$

Examples :

- ▶ Bernoulli model (parameter p)
- ▶ Gaussian model (parameter (μ, σ))
- ▶ Binomial model (parameter (n, p))

Objective

If we assume that the data were generated from some $p_{\theta^*} \in \mathcal{P}$, with a unknown parameter θ^* , our goal is to find a good estimation of θ . If the data are indeed generated by a distribution in $\mathcal{P}$, the problem is said to be **well specified**. Otherwise, the problem is said to be **misspecified**.

Definition

Likelihood

Let $\mathcal{P} = \{p_\theta, \theta \in \Theta\}$ be a parametric model.

Given $y \in \mathcal{Y}$, the **likelihood** is the function :

$$\theta \mapsto L(\theta|y) = p_\theta(y) \quad (13)$$

Given $D_n = (y_1, \dots, y_n)$, the likelihood $L(.|D_n)$ is the function :

$$\theta \mapsto L(\theta|D_n) = \prod_{i=1}^n p_\theta(y_i) \quad (14)$$

The **maximum likelihood estimator** (MLE) is the parameter θ that maximises the likelihood :

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} (L(\theta|D_n)) \quad (15)$$

Remarks

- ▶ Since the samples y_i are assumed to be independent, the likelihood corresponds to the probability (or probability density) of observing the dataset according to p_θ .
- ▶ We often maximise the log of the likelihood, as it is easier to differentiate a sum. Since log is an increasing function, the MLE is also the maximiser of the log of L .

Example 1

Exercise 2: We observe the data $(1, 0)$. We model these data with a Bernoulli distribution of parameter p .

- ▶ What is the likelihood of these observations as a function of p ?
- ▶ What is the value $\hat{p}$ that maximizes this likelihood?

Example 2

Exercise 3: We observe the data $(1, 0, 1)$ (same hypotheses)

- ▶ What is the likelihood of these observations as a function of p ?
- ▶ What is the value $\hat{p}$ that maximizes this likelihood?

Example 3

We observe the data $(2.5, 3.5)$. We assume that these data come from a normal law of parameters μ and σ .

$$\begin{aligned} L &= p(2.5|\mu, \sigma)p(3.5|\mu, \sigma) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{2.5-\mu}{\sigma})^2} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{3.5-\mu}{\sigma})^2} \end{aligned} \quad (16)$$

We want to show that the likelihood is maximum for :

- ▶ $\hat{\mu} = \frac{2.5+3.5}{2}$
- ▶ $\hat{\sigma}^2 = \frac{(2.5-\hat{\mu})^2 + (3.5-\hat{\mu})^2}{2}$

ERM

In the context of density estimation, we can define a loss function as the **negative log-likelihood**.

$$\Theta \times \mathcal{Y} \mapsto -\log(p_{\theta}(y))$$

Given this loss, the risk writes :

$$R(\theta) = E_Y[-\log(p_{\theta}(y))]$$

and the empirical risk :

$$R_n(\theta) = -\frac{1}{n} \sum_{i=1}^n \log(p_{\theta}(y_i))$$

The MLE is then also the empirical risk minimizer.

KL divergence

The Kullback-Leibler divergence is a quantity used to compare two probability distributions.

Definition

Kullback-Leibler divergence

Given two distributions p and q , the KL divergence from p to q is defined as :

$$KL(p||q) = E_{Y \sim p} \left[\log \frac{p(Y)}{q(Y)} \right]$$

Lemma

If the data are generated by p_{θ^} , then $KL(p_{\theta^*}||p_{\theta})$ is the excess risk of p_{θ} , with the negative log-likelihood loss.*

Link with supervised learning methods

Probabilistic modelling can provide an interesting interpretation of several supervised learning methods, such as :

- ▶ logistic regression
- ▶ ordinary least squares

In a supervised learning context, we replace the likelihood $p_{\theta}(y)$ by a **conditional** likelihood $p_{\theta}(y|x)$ (conditional modelling).

$$R_n(\theta) = -\frac{1}{n} \sum_{i=1}^n \log(p_{\theta}(y_i|x_i)) \quad (17)$$

Link with logistic regression

We consider a binary classification problem, with $\mathcal{Y} = \{0, 1\}$.

Let us now consider the probabilistic model such that

$$p_{\theta}(1|x) = \sigma(\theta^T x)$$

Equivalently, this model can be written (remember that $y = 0$ or $y = 1$)

$$p_{\theta}(y|x) = (\sigma(\theta^T x))^y (1 - \sigma(\theta^T x))^{1-y} \quad (18)$$

Exercise 4: Show that the parameter θ with maximum likelihood is the logistic regression estimator θ_{logit} (cross entropy version).

References I